

Lecture 6 (153 - 248)

Nonlinear Regression

$$\textcircled{1} \quad y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t - c) + \underset{\substack{\uparrow \\ \text{error}}}{\varepsilon_t}$$

observed
time series

$$\text{ReLU}(x) = x_+ = \max(x, 0)$$

$c \rightarrow$ unknown

e.g.: $y_t := \log$ population of CA for year t .

$100 \times \beta_1 =$ annual growth rate of CA population before year c

$100 \times (\beta_1 + \beta_2) =$ annual growth rate after year c

$$\textcircled{2} \quad y_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) + \varepsilon_t$$

$\rightarrow$ Sinusoid model

$\rightarrow$ STARTING POINT FOR Spectral Analysis

$$\beta_0 + \sum_{j=1}^m \left[\beta_{1j} \cos 2\pi f_j t + \beta_{2j} \sin(2\pi f_j t) \right] + \varepsilon_t$$

High-dimensional Model.

$$\begin{aligned} & \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t - c_1) \\ & + \beta_3 \text{ReLU}(t - c_2) \\ & + \beta_4 \text{ReLU}(t - c_3) + \dots + \varepsilon_t \end{aligned}$$

Today:

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t-c) + \varepsilon_t$$

Unknown parameters: $c, \beta_0, \beta_1, \beta_2$ $\nearrow \sigma$
 $\varepsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

Point Estimates

& Uncertainty Quantification.

How to estimate $c, \beta_0, \beta_1, \beta_2$?

Least Squares

$$S(\beta_0, \beta_1, \beta_2, c) = \sum_{t=1}^n \left[y_t - \beta_0 - \beta_1 t - \beta_2 (t-c)_+ \right]^2$$

→ Minimize over $\beta_0, \beta_1, \beta_2, c$

$$\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{c}$$

First fix c , then minimize over $\beta = (\beta_0, \beta_1, \beta_2)$

$$\rightarrow \hat{\beta}(c)$$

$$\hat{\beta}(c) = \underset{\text{over } \beta_0, \beta_1, \beta_2}{\text{minimizer of}} \sum_{t=1}^n \left(y_t - \beta_0 - \beta_1 t - \beta_2 \underbrace{(t-c)_+}_{\mathbb{I}(t \geq c)} \right)^2$$

$$\hat{\beta}(c) = (X_c^T X_c)^{-1} X_c^T y$$

$$X_c = \begin{bmatrix} 1 & 1 & (1-c)_+ \\ \vdots & \vdots & \vdots \\ 1 & n & (n-c)_+ \end{bmatrix}$$

$$S(\beta, c) \xrightarrow{\text{minimize over } \beta} S(\hat{\beta}(c), c)$$

$$\hat{c} = \text{minimizer of } S(\hat{\beta}(c), c)$$

Take some range of values of c
 & then do grid search.

$S(\hat{\beta}(c), c) \rightarrow$ Smallest sum of squares
 in the linear regression when
 c is fixed.

||

$$RSS(c)$$

$$\hat{c} = \text{minimizer of } RSS(c)$$

$$y_t = \beta_0 + \beta_1 t + \beta_2 (t - c) + \varepsilon_t$$

$$t = 1, \dots, n \quad c \in (1, n)$$

No need to consider $c \leq 1$ or $c \geq n$.

$$(c \leq 1) \rightarrow \beta_0 + \beta_1 t + \beta_2 (t - c)$$

$$= \beta_0 - \beta_2 c + (\beta_1 + \beta_2) t$$

$\rightarrow$ linear regression model

$$(c \geq n) \quad (\beta_0 + \beta_1 t) \rightarrow \text{linear regression}$$

$$c \in (1, n) \rightarrow c \in (c, n - c)$$

Bayesian Inference

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t-c) + \varepsilon_t$$

$$\varepsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

Likelihood:

$$y_t \stackrel{\text{iid}}{\sim} N(\beta_0 + \beta_1 t + \beta_2 (t-c)_+, \sigma^2)$$

$$\prod_{t=1}^n \left(\frac{1}{\sqrt{2\pi} \sigma} \right) \exp \left[- \frac{(y_t - \beta_0 - \beta_1 t - \beta_2 (t-c)_+)^2}{2\sigma^2} \right]$$

$$= (2\pi)^{-n/2} \sigma^{-n} \exp \left[- \frac{S(\beta, c)}{2\sigma^2} \right] \rightarrow \text{likelihood}$$

$$\beta_0, \beta_1, \beta_2, \text{log} \sigma \stackrel{\text{iid}}{\sim} \text{Unif}(-C, C)$$

$$\rightarrow c \sim \text{unif}(1, n) \text{ or } c \sim \text{unif}(5, n-5)$$

$$c \sim \text{unif}(R)$$

$R \rightarrow$ range given by an interval in $(1, n)$

Priors:

$$f(\beta_0, \beta_1, \beta_2, \sigma, c) = \frac{I(-C < \beta_0, \beta_1, \beta_2, \text{log} \sigma < C)}{(2C)^4 \sigma} \frac{I(1 < c < n)}{(n-1)}$$

$$\beta_0, \beta_1, \beta_2, \sigma, c$$

$$C = \infty \propto \frac{1}{\sigma} I(\sigma > 0) I(1 < c < n) \rightarrow (n-1)$$

posterior (β, σ, c)

$$\propto \sigma^{-n} \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) \times \frac{1}{\sigma} \frac{I(\sigma > 0)}{I(1 < c < n)}$$

posterior (c)

$$\propto \int \int \sigma^{-n} \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) \frac{1}{\sigma} \frac{I(\sigma > 0)}{I(1 < c < n)} d\beta d\sigma$$

$$= I(1 < c < n) \int_0^\infty \int \sigma^{-n-1} \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) d\beta d\sigma$$

$$= I(1 < c < n) \int_0^\infty \sigma^{-n-1} \left[\int \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) d\beta \right] d\sigma$$

$$S(\beta, c) = \sum_t \left[y_t - \beta_0 - \beta_1 t - \beta_2 (t-c)_+ \right]^2$$

↑
squares
function

$$= \|y - X_c \beta\|^2$$

Pythagorean
Identity

$$= \|y - X_c \hat{\beta}_c + X_c \hat{\beta}_c - X_c \beta\|^2$$

$$= \|y - X_c \hat{\beta}_c\|^2 + \|X_c \hat{\beta}_c - X_c \beta\|^2$$

$$= \text{RSS}(c) + \|X_c (\beta - \hat{\beta}_c)\|^2$$

$$= \text{RSS}(c) + (\beta - \hat{\beta}_c)^T X_c^T X_c (\beta - \hat{\beta}_c)$$

$$S(\beta, c) = \text{RSS}(c) + (\beta - \hat{\beta}_c)^T X_c^T X_c (\beta - \hat{\beta}_c)$$

$$\int \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) d\beta$$

$$= \int \exp\left(-\frac{\text{RSS}(c)}{2\sigma^2}\right) \exp\left(-\frac{(\beta - \hat{\beta}_c)^T X_c^T X_c (\beta - \hat{\beta}_c)}{2\sigma^2}\right) d\beta$$

$$= \exp\left(-\frac{\text{RSS}(c)}{2\sigma^2}\right) \int \exp\left(-\frac{(\beta - \hat{\beta}_c)^T X_c^T X_c (\beta - \hat{\beta}_c)}{2\sigma^2}\right) d\beta$$

$p = \text{dimension of } \beta$

$$\int \exp\left(-\frac{(\beta - \mu)^T \Sigma^{-1} (\beta - \mu)}{2}\right) d\beta$$

$$= (2\pi)^{\frac{p}{2}} \sqrt{\det \Sigma}$$

$$\mu = \hat{\beta}_c$$

$$\Sigma^{-1} = \frac{X_c^T X_c}{\sigma^2}$$

$$\Rightarrow \Sigma = \sigma^2 (X_c^T X_c)^{-1}$$

$$= \exp\left(-\frac{\text{RSS}(c)}{2\sigma^2}\right) (2\pi)^{\frac{p}{2}} \sqrt{\det(\sigma^2 (X_c^T X_c)^{-1})}$$

$$= \exp\left(-\frac{\text{RSS}(c)}{2\sigma^2}\right) (2\pi)^{\frac{p}{2}} \sigma^p |X_c^T X_c|^{-1/2}$$

$$\det(cA) = c^p \det(A)$$

$p \times p$

$$|A| = \det A$$

Recap: $c \sim \text{Unif}(1, n)$

posterior (β, σ, c)

$$\text{posterior}(c) = \int \int \text{post}(\beta, \sigma, c) d\beta d\sigma$$

$$\propto I(1 < c < n)$$

$$\int_0^\infty \sigma^{-n-1} \exp\left(-\frac{\text{RSS}(c)}{2\sigma^2}\right) (2\pi)^{p/2} \sigma^p \left(X_c^T X_c\right)^{-1/2} d\sigma$$

$$\propto I(1 < c < n) |X_c^T X_c|^{-1/2}$$

$$\int_0^\infty \sigma^{-n+p-1} \exp\left(-\frac{\text{RSS}(c)}{2\sigma^2}\right) d\sigma$$

$$\propto I(1 < c < n) |X_c^T X_c|^{-1/2}$$

$$\left[\frac{1}{\text{RSS}(c)}\right]^{\frac{n-p}{2}}$$

$$\int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) d\sigma \propto \left(\frac{1}{S(\beta)}\right)^{n/2}$$

posterior (c)

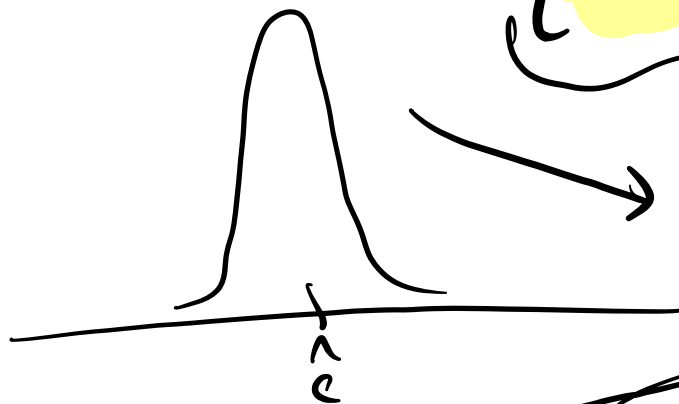
$$\propto I(1 < c < n) |X_c^T X_c|^{-1/2}$$

$$\left(\frac{1}{\text{RSS}(c)}\right)^{\frac{n-p}{2}}$$

$$X_c = \begin{pmatrix} 1 & 1 & (1-c)_+ \\ \vdots & 2 & \vdots \end{pmatrix}$$

$\hat{c}$: least squares estimator
 $\rightarrow$ minimizer of $RSS(c)$

posterior (c)
 $\propto I(1 < c < n) |X_c^T X_c|^{-1/2} \left[\frac{RSS(\hat{c})}{RSS(c)} \right]^{\frac{n-p}{2}}$



highly concentrated around $\hat{c}$.

$\left(\frac{RSS(\hat{c})}{RSS(c)} \right)^{\frac{n-p}{2}}$